Impact of Multi-View Fusion and Biomechanical Modeling on Markerless Motion Tracking

Review – 20/11/2025

Key Hypotheses

1 - Impact of biomechanical modelisation

Biomechanical model ➔

= <u>**Hard** joint constraints</u>

OpenSim

SMPL model (Skinned Multi-Person Linear model)

= <u>**Soft** joint constraints</u>

SMPL model VS SMPL model + **OpenSim**

Only on **single-view** methods !!

2 - Multi-view vs Single-view

- 13 single-view methods
- 2 multi-view methods

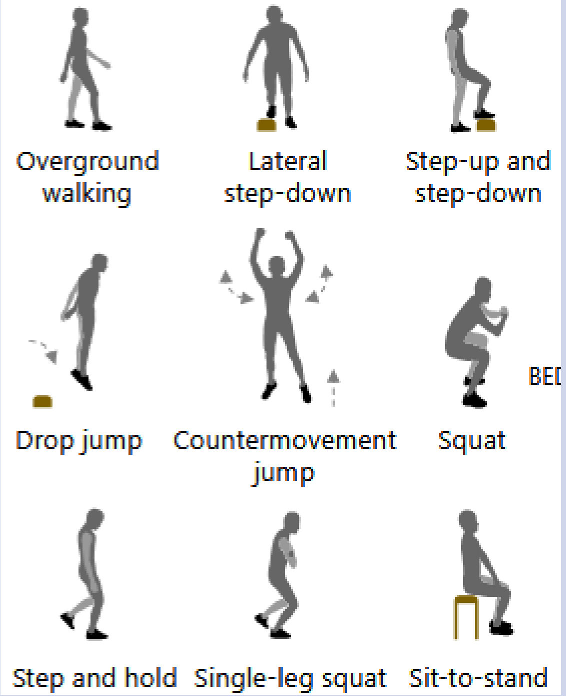
Theida 3D

OpenCap (≈ MOCAPIA)

What gains justify switching technology?

DATA process

23 participants – controlled indoor environments



Markerless Solution

- 20 RGB cameras
- 100Hz
- Calibration board

Marker Solution (reference)

- 20 IR cameras
- 100Hz
- 45 physical markers
- A wand calibration

Single-View methods

- All of them are based the SMPL model

Multi-View methods

OpenCap	Theida 3D

- No SMPL model
- But a IK correction with OpenSim

Tools of evaluation

➔**RMSD** (Root Mean Square Difference)

A biomechanical tool

➔**PA-MPJPE**

A computer-vision tool

➔ **Linear Mixed-Effects Model** + **ANOVA III**

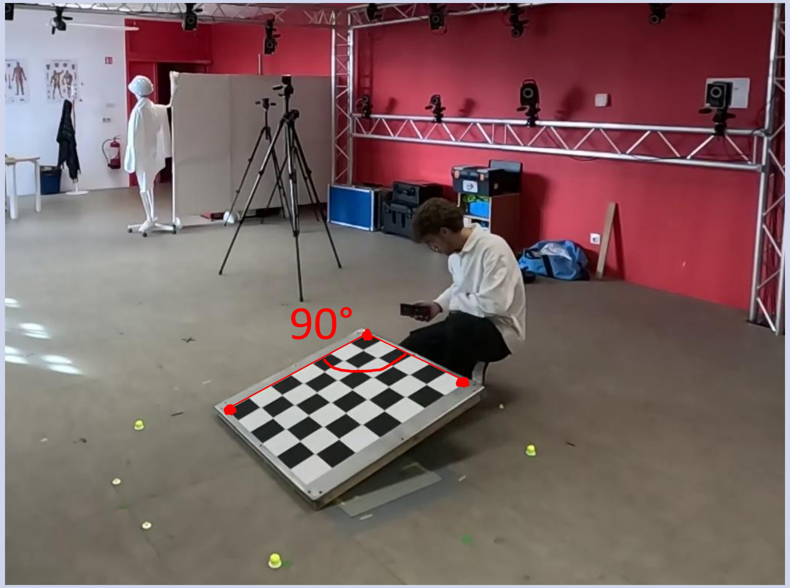
<u>LMEM</u>: split the fix effects(motion capture methods) and random effects (size, weight of the subject)

<u>ANOVA</u>: evaluate if the result (RMSD) is statistically true ($p < 0{,}05$)

The process

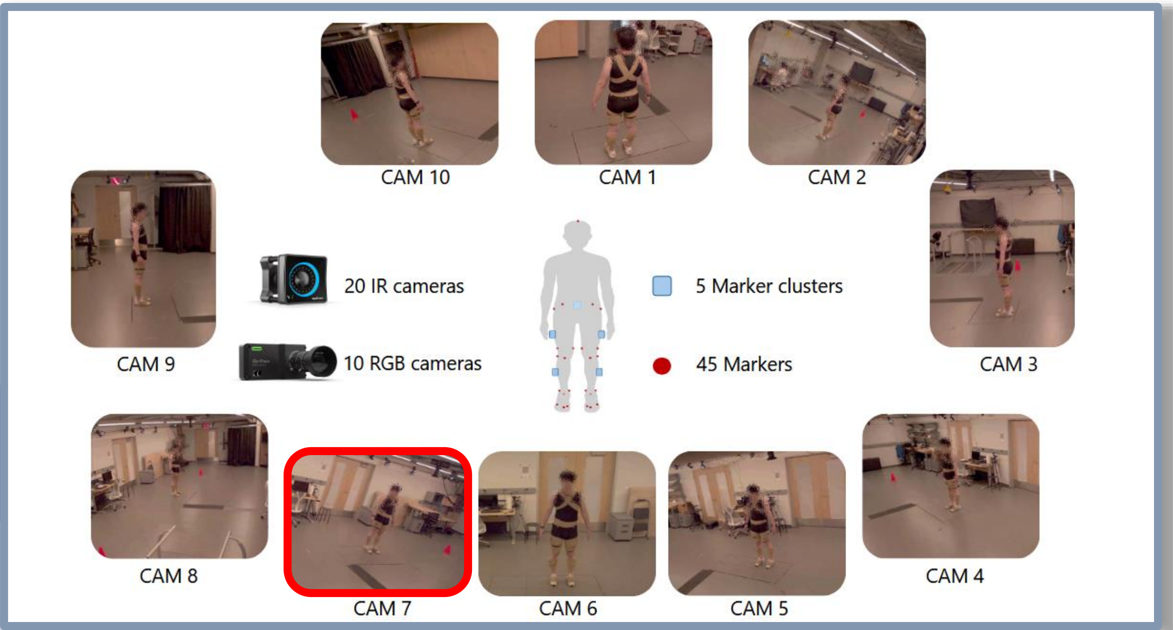
Check of the Single-View Calibration

- RMSE of the corner of the board < 2°



They didn’t use reprojection error

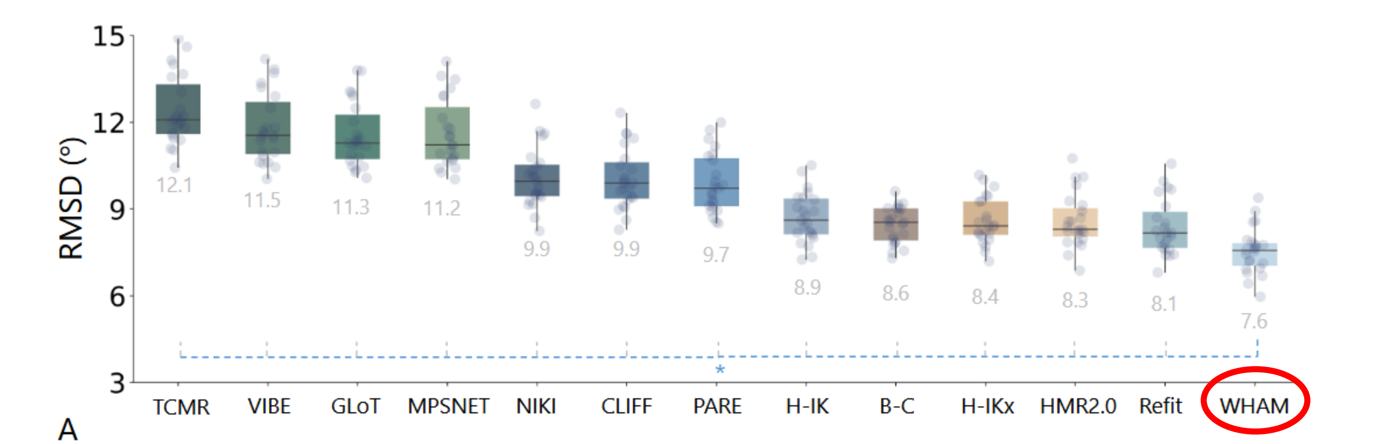
Process 2/4 – Primary evaluation



Evaluation of 13 single view methods

- Evaluated across all 10 cameras – 33Hz (same as us)

➔ Average RMSD across all views after a Kruskal-Wallis test

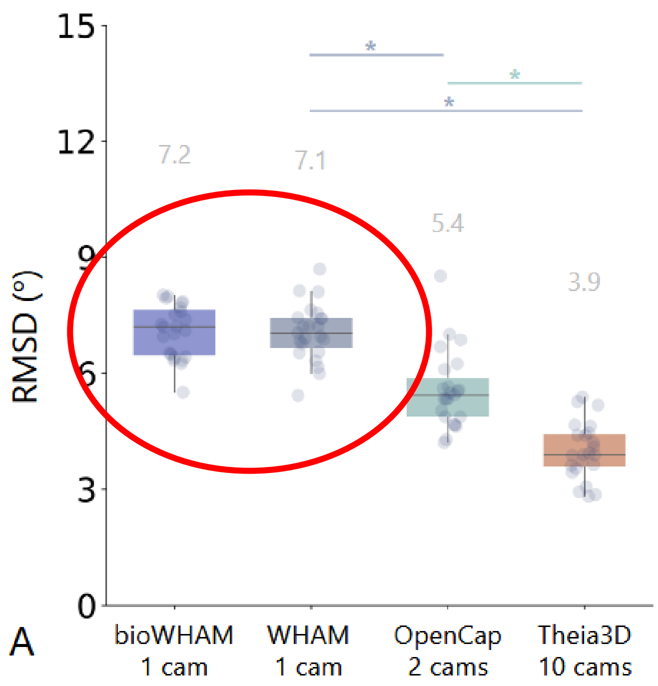


➔ WHAM achieved the lowest RMSD with 7.6°

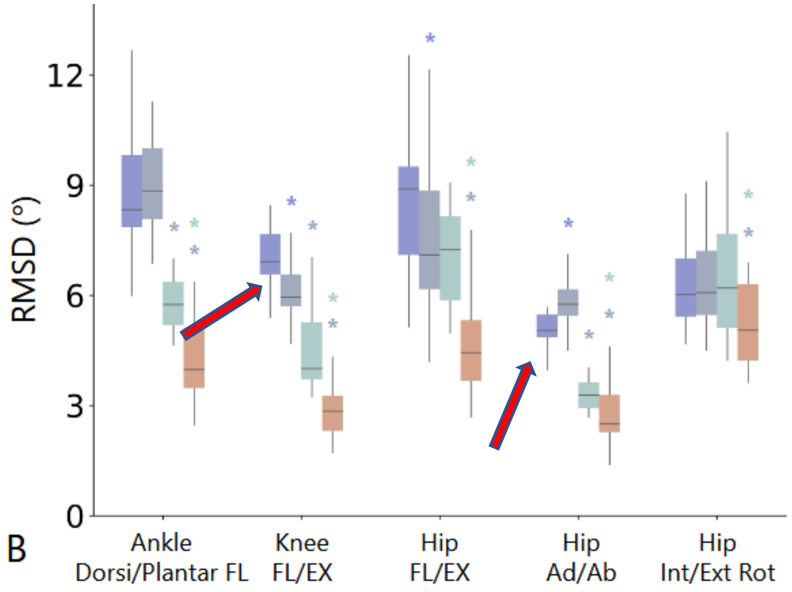
Results- hypothese 1

WHAM model (camera 7) → Use of the **SMPL** model on WHAM → WHAM with «soft» constraints

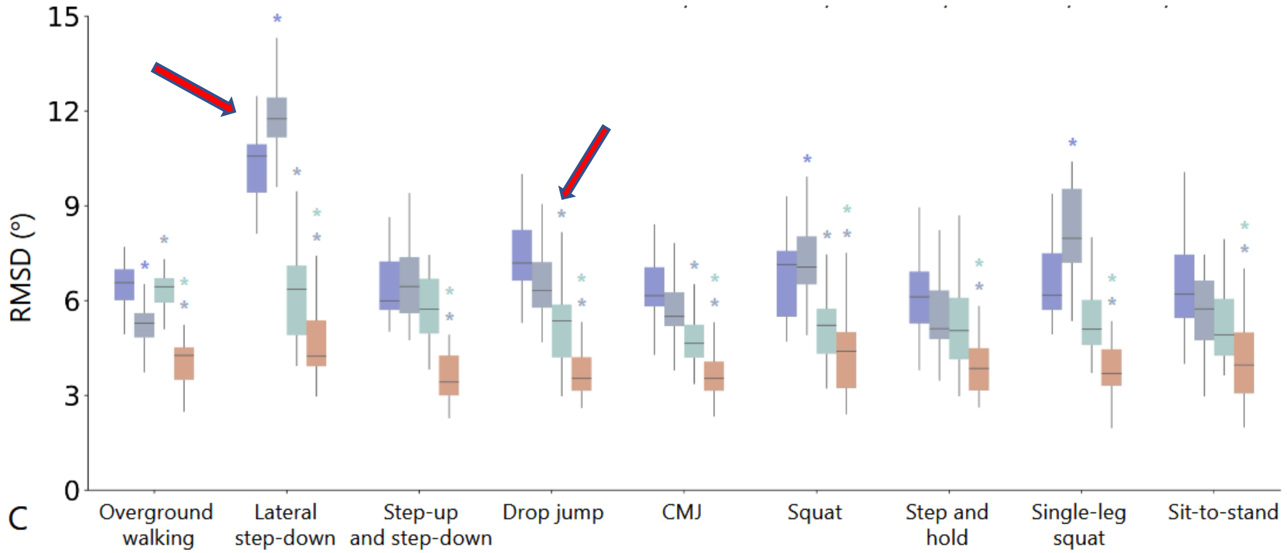
WHAM model (camera 7) → Use of the **SMPL** model on WHAM → 29 markers used for OpenSim IK → **bioWHAM**



A



B

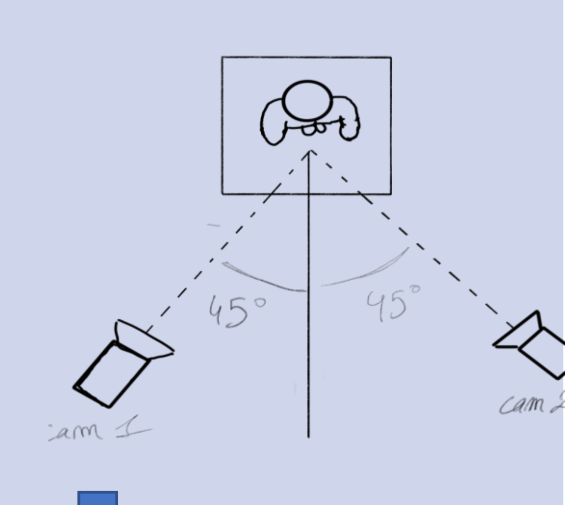


C

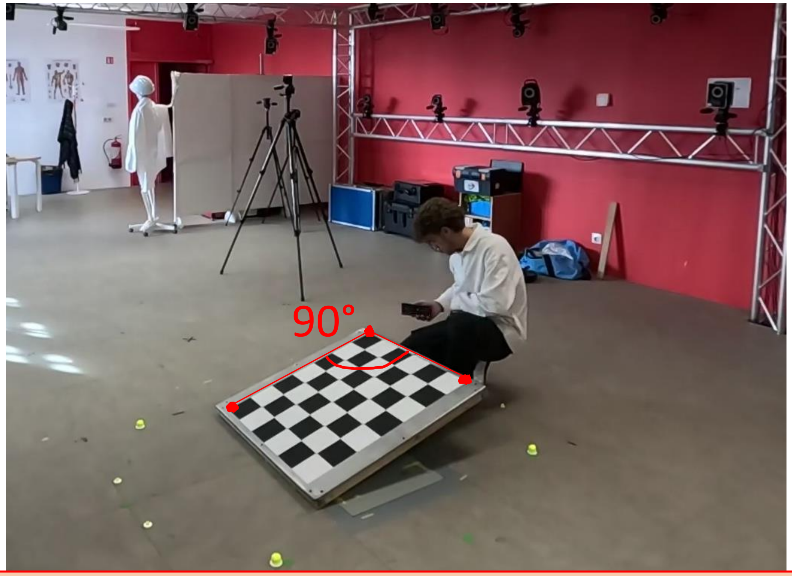
Process 4/4 – Hypothese 2

OpenCap

2 cameras



Calibration



The same calibration is used for OpenCap et Theia3D – no reprojection error study

No accuracy gains by adding cameras

Estimateur de pose 2D

HRNet	OpenPose

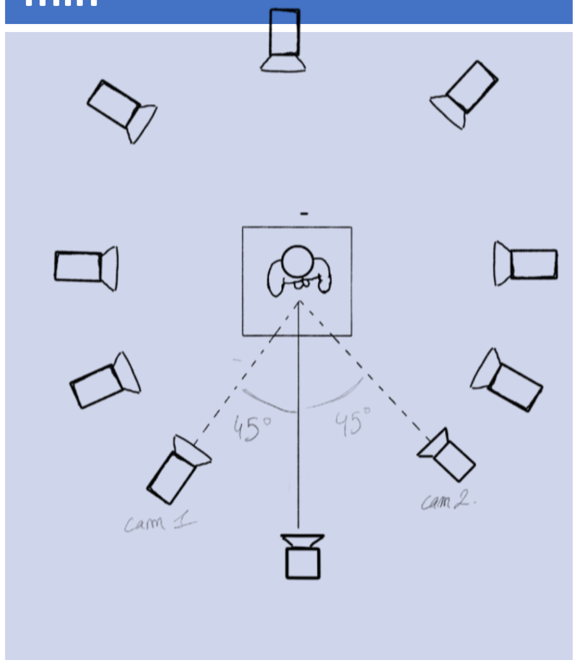
Considered less accurate than **RTMPose** (used in **MoCapIA**)

triangulation → filtering → Marker Augmentation → IK on OpenSim

Results– Hypothese 2

Theia3

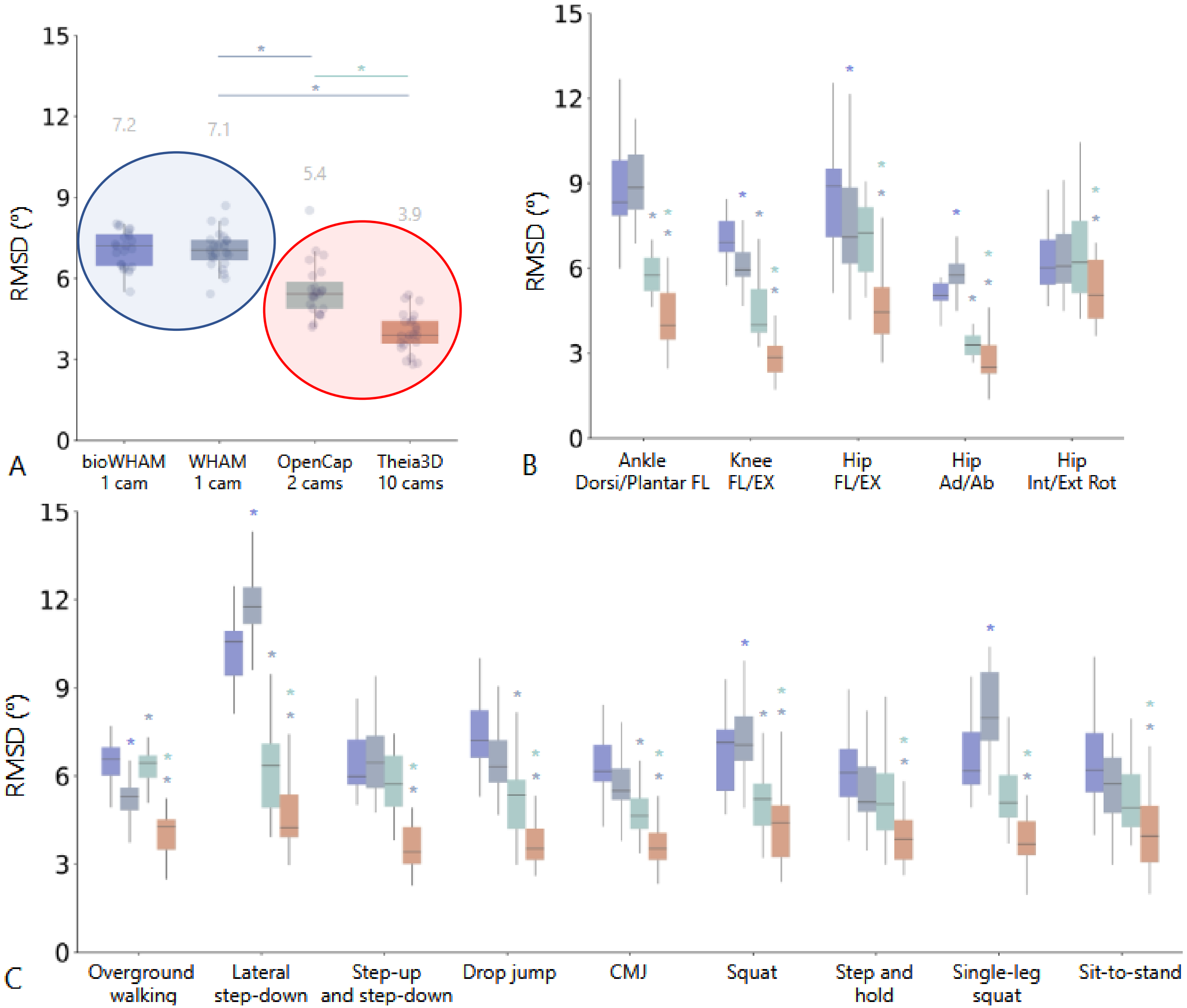
10 cameras – 6 cameras min



Black-Box (private licence)

WHAM/bioWHAM

OpenCap



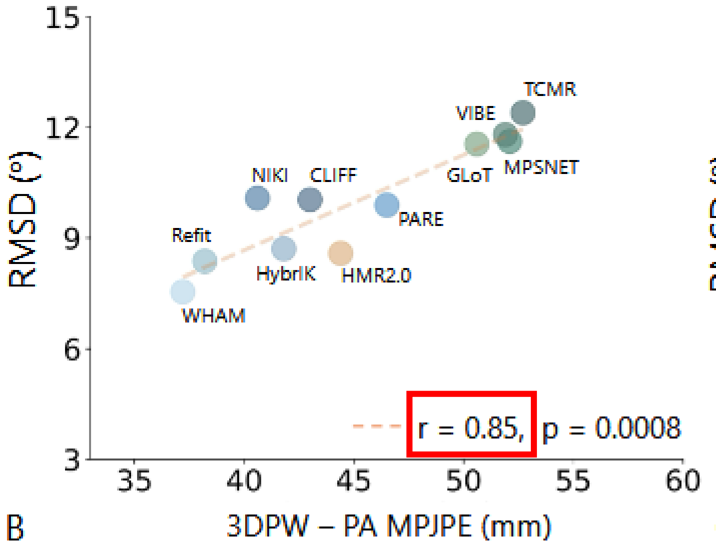
Auxiliary Finding

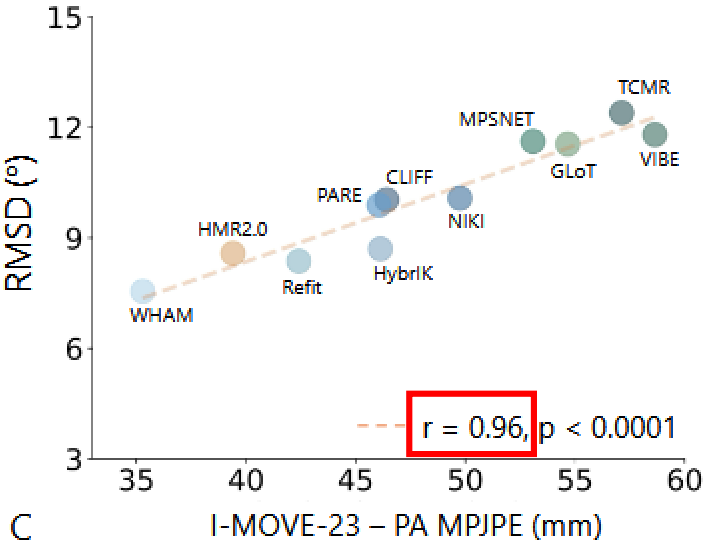
Datasets	
3DPW	**I-MOVE-23**
Standart reference dataset in CV	data collected for this study

Correlation between **RMSD** and **PA-MPJPE**

An metric used in the field of **biomechanic**

An metric used in the field of **computer vision**





Conclusion

Both metrics are **highly correlated**

Both of them could be used in MoCapIA’s project